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Stack-Intro

Stack

- It is a "pile of objects"

- It is a linear data-structure that supports insertion (push) and deletion (pop) from the same end (top)

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- L17 - Pointers and Character...
- L18 - Character Arrays and S...
- L19 - Strings and 2D Arrays
- L20 - 2D Arrays
- L21 - 2D Arrays and DMA
- L22 - Algo. Analysis and Bina...
- L23 - Binary Search
- L24 - OOP
- L25 - Recursion
- L26 - Recursion
- L27 - Recursion
- L28 - Recursion
- L29 - Recursion
- L30 - Recursion
- L31 - Linked List
- L32 - Linked List
- L33 - Linked List
- L34 - Stacks

Add section

- Stack-Intro
- Stack-Insert At Bottom
- Stack-Reverse
- Valid Parentheses
- Min-Stack
- Asteroid Collision
- [HW] Iterative Tower of Hanoi
- Homework

Add page

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Add page

Homework

https://leetcode.com/problems/decode-string/
https://leetcode.com/problems/evaluate-reverse-polish-notation/
https://leetcode.com/problems/basic-calculator-ii/
https://leetcode.com/problems/basic-calculator/

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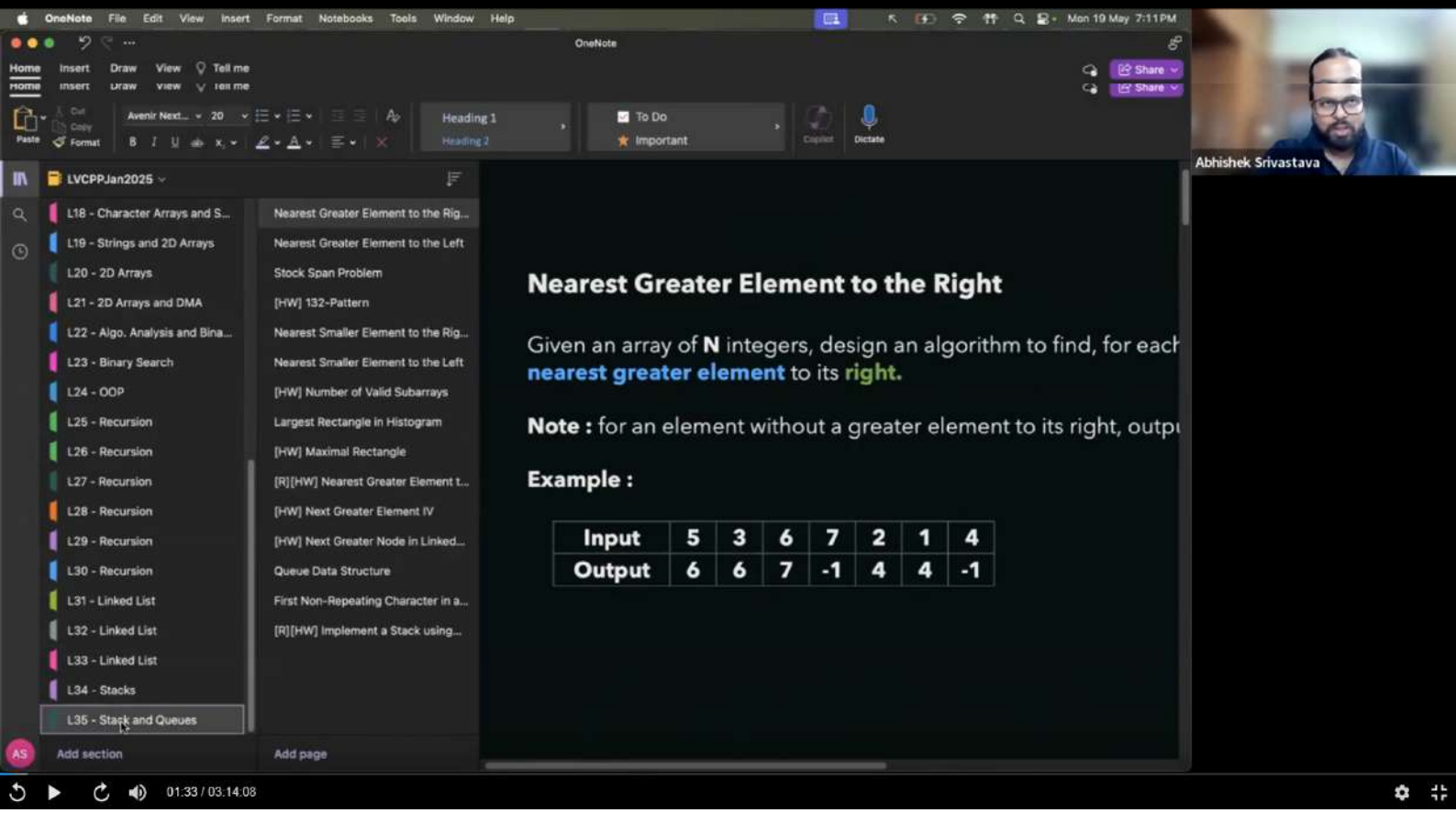
Nearest Greater Element to the Right

Given an array of **N** integers, design an algorithm to find, for each element, the **nearest greater element** to its **right.**

- **Note :** for an element without a greater element to its right, output **-1.**

Example :

Input	**5**	**3**	**6**	**7**	**2**	**1**	**4**
Output	**6**	**6**	**7**	**-1**	**4**	**4**	**-1**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L23 - Binary Search
L24 - OOP
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L26 - Recursion
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L28 - Recursion
L29 - Recursion
L30 - Recursion
L31 - Linked List
L32 - Linked List
L33 - Linked List
L34 - Stacks
L35 - Stack and Queues
Add section
Nearest Greater Element to the Rig...
Nearest Greater Element to the Left
Stock Span Problem
[HW] 132-Pattern
Nearest Smaller Element to the Rig...
Nearest Smaller Element to the Left
[HW] Number of Valid Subarrays
Largest Rectangle in Histogram
[HW] Maximal Rectangle
[R][HW] Nearest Greater Element t...
[HW] Next Greater Element IV
[HW] Next Greater Node in Linked...
Queue Data Structure
First Non-Repeating Character in a...
[R][HW] Implement a Stack using...
Add page
Nearest Greater Element to the Right
Given an array of N integers, design an algorithm to find, for each
nearest greater element to its right.
Note : for an element without a greater element to its right, outpu
Example :
Input 5 3 6 7 2 1 4
Output 6 6 7 -1 4 4 -1
Abhishek Srivastava
01:33 / 03:14:08

Nearest Greater Element to the Left

Given an array of **N** integers, design an algorithm to find, for each element, the **nearest greater element** to its **left.**

Note : for an element without a greater element to its left, output **-1.**

Example :

Input	**5**	**2**	**0**	**4**	**1**	**3**	**6**
Output	**-1**	**5**	**2**	**5**	**4**	**4**	**-1**

Stock Span Problem

Given prices of a stock for **N** consecutive days, design an algorithm to find, for each day, **span of stock price** on that day.

We define **span of stock price today** as the **maximum** number of consecutive days (staring from today and going backwards) for which the stock price was **less than or equal** to today's price.

Example :

n=7

	0	1	2	3	4	5	6
Input	100	80	60	70	60	75	85
Output	1	1	1	2	1	4	6

Stock Span Problem

Given prices of a stock for **N** consecutive days, design an algorithm to find, for each day, **span of stock price** on that day.

We define **span of stock price** **today** as the **maximum** number of consecutive days (staring from today and going backwards) for which the stock price was **less than or equal** to today's price.

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Example :

n=7

	0	1	2	3	4	5	6
Input	**100**	**80**	**60**	**70**	**60**	**75**	**85**
Output	**1**	**1**	**1**	**2**	**1**	**4**	**6**

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Add section

- Nearest Greater Element to the Rig...
- Nearest Greater Element to the Left
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Add page

[HW] 132-Pattern

132-Pattern

Given an array of **N** integers **nums**, design an algorithm to check
132-Pattern in **nums**. We define a **132-Pattern** as a subsequen
nums[i], nums[j] and nums[k] such that

- **i < j < k** and
- **nums[i] < nums[k] < nums[j].**

Example :

N = 6

	0	1	2	3	4	5
Input	**3**	**4**	**6**	**2**	**1**	**5**

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Nearest Smaller Element to the Right

Given an array of **N** integers, design an algorithm to find, for each element, the **index** of **nearest smaller element** to its **right.**

Note : for an element without a smaller element to its right, output **N**

Example :

	0	1	2	3	4	5	6
Input	2	5	1	3	6	4	0
Output	2	2	6	6	5	6	7

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Nearest Smaller Element to the Left

Given an array of **N** integers, design an algorithm to find, for each element, the **index** of **nearest smaller element** to its **left.**

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Note : for an element without a smaller element to its left, output **-1**

Example :

	0	1	2	3	4	5	6
Input	0	3	5	4	1	6	2
Output	-1	0	1	1	0	4	4

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Number of Valid Subarrays

Given an integer array **nums**, design an algorithm to find the number of non-empty sub-arrays with the leftmost element of the subarray not larger than other elements in the subarray.

Example :

	0	1	2	3	4	5
Input	1	4	2	5	3	9

Output : 15